Review Article

Fracture-Related Infection—Epidemiology, Etiology, Diagnosis, Prevention, and Treatment

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Zusammenfassung

Background: Fracture-related infection (FRI) is a challenge to physicians and other workers in health care. In 2018, there were 7253 listed cases of FRI in Germany, corresponding to an incidence of 10.7 cases per 100 000 persons per year.

Methods: This review is based on pertinent publications retrieved from a search in PubMed with the search terms "fracture," "infection," "guideline," and "consensus." Aside from the primary literature, international guidelines and consensus recommendations were evaluated as well.

Results: FRI arise mainly from bacterial contamination of the fracture site. Staphylococcus aureus is the most commonly detected pathogen. The treatment is based on surgery and antibiotics and should be agreed upon by an interdisciplinary team; it is often difficult because of biofilm formation. Treatment options include implant-preserving procedures and single-stage, two-stage, or multi-stage implant replacement. Treatment failure occurs in 10.3% to 21.4% of cases. The available evidence on the efficacy of various treatment approaches is derived mainly from retrospective cohort studies (level III evidence). Therefore, periprosthetic joint infections and FRI are often discussed together.

Conclusion: FRI presents an increasing challenge. Preventive measures should be optimized, and the treatment should always be decided upon by an interdisciplinary team. Only low-level evidence is available to date to guide diagnostic and treatment decisions. High-quality studies are therefore needed to help us meet this challenge more effectively.

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Periprosthetic joint infections and fracture-related infections (FRI) are becoming increasingly important in the field of orthopedic trauma surgery, with 16 174 and 7253 inpatient cases, respectively, treated in 2018 (1). The rising number of surgical fracture treatments that are needed and that involve the implantation of osteosynthesis materials, combined with the growing proportion of older people represent a challenge for the current and future care of patients (2). Periprosthetic joint infections and FRI are often dealt with together. The complication posed by infections in the fracture region was only recognized in the international literature in 2018 with the introduction and definition of the specific term 'fracture-related infection' (3).

Methods

This review is based on a selective literature search in PubMed with the search terms "fracture AND infection AND guidelines," as well as "fracture AND infection AND consensus." In addition to international guidelines and consensus recommendations, the primary literature was also evaluated. The aim of this study was to formulate current standards in diagnosis and treatment in the form of practical recommendations and to familiarize the reader with key aspects of epidemiology, etiology, and prevention.

Epidemiology

In 2018, 1.23% of all fractures treated in Germany on aninpatient basis resulted in an FRI (1). Between 2008 and 2018, a slight increase in administrative incidence from 5556 cases

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Table 1

Confirmatory and suggestive criteria of a fracture-related infection according to Metsemakers et al. (3)

Confirmatory criteria	Suggestive criteria
Fistula, sinus or wound dehiscence (connected to the bone or implant)	Clinical symptoms – Pain (without weight-bearing, worsening over time, new-onset) – Local redness – Local swelling – Elevated local temperature – Fever (≥ 38.3° C)
Detection or discharge of pus from the wound	Radiological signs – Bone lysis (at the fracture site or around the implant) – Implant loosening – Sequestration – Failure of bone to heal (e.g., nonunion) – Presence of periosteal bone formation (e.g., at sites other than the fracture site or in the case of a consolidated fracture)
Phenotypically indistinguishable pathogens, identified by culture from at least two separate deep tissue/implant specimens	A pathogenic organism identified by culture from a single specimen taken from deep tissue/implant
Presence of microorganisms, confirmed through histopathological analysis with specific staining techniques for bacteria or fungi, or more than five polymorphonuclear leukocytes per high-power field	Elevated inflammatory markers – Erythrocyte sedimentation rate (ESR) – White blood cell count – C-reactive protein (CRP)
	Persistent, increasing, or new-onset wound drainage beyond the initial postoperative days for which there is no solid alternative explanation
	New-onset joint effusion

(8.4/100 000 persons) to 7253 cases (10.7/100 000 persons) was observed in Germany (1). One can assume that the worldwide incidence of FRI will continue to rise (e1). FRI poses a particular challenge in countries with medium and low average incomes, since here not only is the incidence higher due to the greater number of open fractures but resources for surgical or antibiotic treatment is also limited (e2).

Etiology

FRI arise mainly from bacterial contamination of the fracture site. In the case of open fractures, this can occur as a result of the trauma itself, while in the case of closed fractures, this may also result from inoculation of pathogens into the wound area during surgery. Overall, the risk of FRI increases with the severity of soft tissue damage. Whereas an infection rate of 1–2% is assumed for closed tibial-shaft fractures, this rate rises to 42.9% for open fractures with extensive soft tissue injury (Gustilo–Anderson type III) (4, 5). Hematogenous infection—the colonization of pathogens originating in the skin, oral cavity, and respiratory or urinary tract via the blood to the fracture region—is less common (5). Biofilm formation of the infection-causing pathogen poses a challenge. Biofilm refers to the aggregation of freely moving (= planktonic) bacterial cells on surfaces. Once there, they become sessile cells forming an extracellular polysaccharide matrix (e3). Bacteria in the planktonic phase are considered to be responsible for a more acute infection process, but are on the whole readily treatable, while the sessile form is associated with a slower infection process and less favorable treatment options.

Pathogen spectrum

As in other musculoskeletal infections, *Staphylococcus aureus* (31.4–37.4%) and *Staphylococcus epidermidis* (16.9–25.8%) are the most common FRI-causing pathogens. Other staphylococci (8.4–18%), streptococci (7.2%), enterococci (2.4%), and *Cutibacterium* species (2.4%) represent additional gram-positive pathogens, whereas gram-negative bacteria, in particular *Enterobacteriaceae* and *Pseudomonas* species, account for approximately one fifth of all pathogens (20.5–23%) (6–8). The rate of polymicrobial infections varies (8.6–36%), with these occurring more frequently in acute infections (7–9). The pathogen spectrum does not appear to be influenced by whether the FRI is an acute or chronic infection (9). Antibiotic-resistant pathogens play an important role worldwide. Although methicillin-resistant *Staphylococcus aureus* (MRSA) is only a marginal occurrence in FRI in Germany (~ 1%), it is of greater relevance in the USA (44.1%) and China (25.3%) (e4–e6). Therefore, periods abroad should be taken into account when taking a patient's history of MRSA risk.

Definition and diagnosis

As a result of the previous lack of a definition, diagnostic criteria for periprosthetic infections were often extrapolated (e7). A consensus definition for FRI, including both confirmatory and suggestive diagnostic criteria, was finally published in 2018 *(Table 1)* (3).

Diagnosis is to a great extent clinical routine. In addition to FRI itself, consideration should also be given to systemic diseases, and these should be discussed and treated with specialist colleagues in an interdisciplinary

exchange *(Figure 1)*. In order to optimize microbiological diagnosis, it is recommended to take between three and five tissue specimens instead of the usual intraoperative swabs. These should each be taken with a separate sterile instrument from the infected region and not from the area of the skin or fistula. A possible complementary diagnostic measure is to additionally send the implant for sonication (e8). Here, the foreign material as a whole is subjected to ultrasound treatment in order to dislodge any bacteria from the surface of the material and from the biofilm.

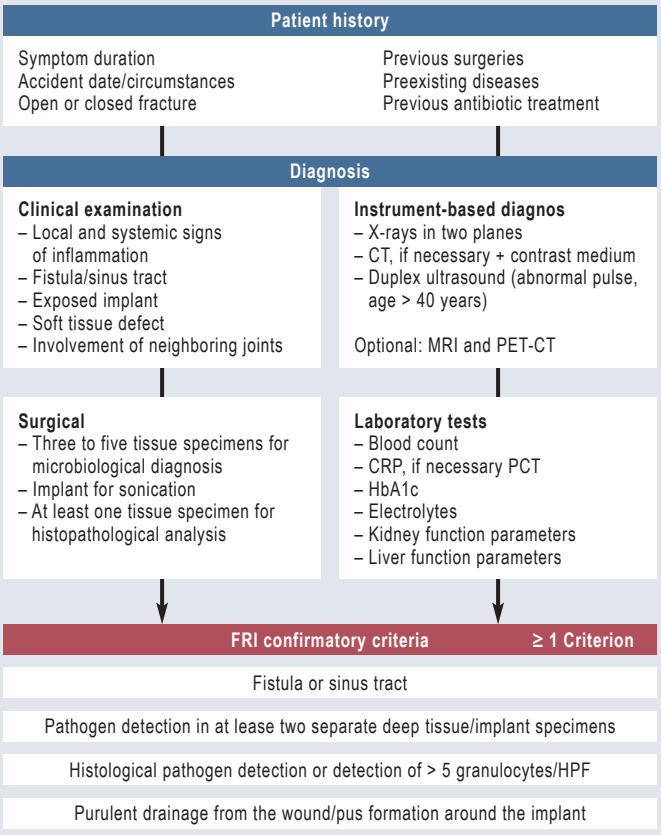
Preventive measures

In addition to general measures recommended by the World Health Organization (WHO) and the Centers for Disease Control and Prevention (10, 11), such as hand disinfection, use of sterile instruments, and repeated sterile cleansing of the surgical site, recent studies show that alcohol-based antiseptic solutions such as chlorhexidine for skin disinfection are superior in terms of minimizing the risk of postoperative infections (11, 12).

The co-treatment of comorbidities such as underlying cardiac disease, peripheral arterial occlusive disease (PAOD), and type II diabetes mellitus is an important factor (13, 14). Moreover, perioperative antibiotic prophylaxis is essential in orthopedic and trauma surgical procedures. In the case of closed fractures and elective procedures, a single dose of a first-generation cephalosporin, for example, cefazolin, 15–60 min before the start of surgery is recommended. If surgery time exceeds 2–3 h, redosing should take place, as has been shown in retrospective cohort studies (15, 16). With regard to open fractures, the evidence on type and duration of antibiotic prophylaxis is less consistent (15). In these cases, intravenously administered antibiotic prophylaxis should be carried out as early as possible (e9). In the case of grade I and II open fractures according to Gustilo–Anderson, first- and second-generation cephalosporins or aminopenicillins plus beta lactamase inhibitors are recommended, although their administration beyond 24 h is not recommended (17). For grade III open fractures, the microbial spectrum of gram-negative bacteria needs to be more broadly covered, hence piperacillin/tazobactam are the drugs of choice. Antibiotic prophylaxis for more than 72 h was not superior in terms of reducing the rate of fracture-related infections in grade III open fractures (18). In complex and high-grade open fractures, a beneficial effect has been demonstrated for the local application of antibiotics. This can take the form of polymethyl methacrylate (PMMA) beads, collagen fleeces, or bone regeneration materials in combination with various antibiotics (19). In open fractures with severe Gustilo–Anderson type IIIB soft tissue defects, soft tissue coverage should ideally be carried out within 72 h, since this can significantly reduce infection rates as well as flap failure compared to delayed free-flap reconstruction after 72 h, as shown in a meta-analysis based on data from 35 case series and eight case reports (20).

In surgical treatment, particularly in the case of open fractures, adequate debridement involving the removal of heavily contaminated and necrotic tissue is essential. In addition, thorough wound irrigation with saline solution should be performed in the case of open fractures (21). For

Figure 1



Diagnostic approach in fracture-related infection
CT, computed tomography; CRP, C-reactive protein; FRI, fracture-related infection; HbA1c, hemoglobin A1c; HPF, high power field; MRI, magnetic resonance imaging; PCT, procalcitonin; PET-CT, positron emission tomography-computed tomography

Table 2

Evidence table for various treatment measures in fracture-related infection*

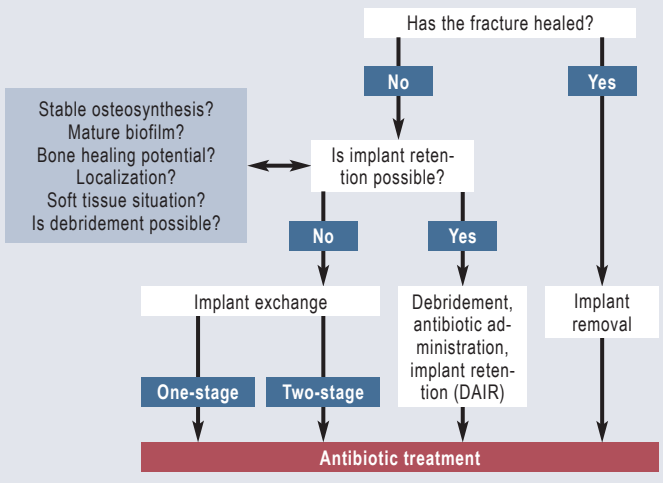
First author, year	Study design/intervention	Follow-up	Outcome	Evidence level according to (34)
Rupp 2023 (25)	Study with comparison group: multidisciplinary treatment (n = 59) compared to control group without multidisciplinary treatment (n = 58)	12 Months	– Amputations: 3.4% versus 6.8%, p = 0.014 – Revisions: 1.5 ± 1.2 (0–5) versus 2.2 ± 1.2 (0–7), p = 0.048	III, retrospectively
Patterson 2023 (35)	Study with comparison group: antibiotic bead pouch (n = 59) compared to VAC treatment (n = 41)	33 Months	Low amputation rate and revision surgery rate with antibiotic bead pouch: β = –1.08, 95% CI: [–2.00; –0.17], p = 0.02	III, retrospectively
Sliepen 2022 (29)	Study with comparison group: local antibiotic therapy (n = 251) compared to control group without local antibiotic therapy (n = 182)	At least 12 months	– 25/251 (10%) Infection recurrence with local antibiotic administration – 34/182 (18.7%) Infection recurrence without local antibiotic therapy	III, retrospectively
Sweere 2022 (31)	Study with comparison group: VAC treatment (n = 99) compared to control group without VAC treatment (n = 164)	At least 12 months	28/99 (28.3%) Infection recurrence with VAC treatment versus 19/164 (11.6%) Infection recurrence without VAC treatment, p = 0.013, OR 1.036 [1.008; 1.066]	III, retrospectively
Buijs 2022 (36)	Study without comparison group: cohort study n = 114 acute FRI treated with DAIR	12 Months 23 Months	– Failure: 13% at 12 months – Failure: 18% at 23 months	III, retrospectively
McNally 2022 (30)	Study without comparison group: cohort study n = 433, various treatment strategies	26 Months	– Failure after DAIR 21.4% – Failure after one-step exchange 12.5% – Failure after conversion to external fixation 10.3% (aHR: DAIR versus Ext Fix 2.38 [0.96; 5.73]	III, retrospectively
Rupp 2020 (37)	Study without comparison group: n = 42 infected nonunions treated with multi-step approach	–	– 6/42 (14.3%) Polymicrobial infection – 21/42 (50%) Germ change in revision surgeries – Gram-negative infection only in polymicrobial infections and germ changes	IV, retrospectively
Haidari 2021 (32)	Systematic review Infection rates in VAC treatment Eight studies (prospective and retrospective cohort studies, case control studies), n = 332	–	2.8–34.9% Infection recurrence	III
Morgenstern 2021 (26)	Systematic review Success rates depending on time-related classification Six studies (RCT, retrospective and prospective studies), n = 276	–	Success rates following DAIR treatment: – Acute FRI: 86–100% – Delayed FRI: 82–89% – Late FRI: 67%	III
Kadhim 2017 (38)	Systematic review/meta-analysis Bone healing with various reconstruction techniques 24 Studies (retrospective), n = 504	At least 18 months	Overall bone healing rate 97.3% (95.3–99.1%)	III

*Evidence level III includes case control studies, retrospective cohort studies, and systematic reviews, while evidence level IV includes case series. The results are given in percentages or mean and standard deviation.
aHR, adjusted hazard ratio; DAIR, debridement, antibiotics, and implant retention; Ext Fix, external fixator; FRI, fracture-related infection; OR, odds ratio; CI, confidence interval; RCT, randomized controlled trial; VAC treatment, vacuum-assisted closure treatment

the primary treatment of high-grade open fractures, temporary external fixation by means of fixators is also recommended (22). Preclinical studies have shown a benefit in terms of reducing infection for the antimicrobial coating of osteosynthesis material with antibiotics or silver compared with standard uncoated implants (e10, e11). Coated implants are also already in clinical use. However, the evidence for their benefits is currently still based on case series and reports (evidence level IV) (23, e12–e14).

For postoperative wound care, the general measures such as the use of sterile dressing material and strict adherence to hand hygiene apply. A meta-analysis based on data from three randomized controlled trials was unable to show a difference in infection rates between dressing changes carried out early (< 48 h postoperatively) and in a delayed manner (> 48 h postoperatively) (24). Nevertheless, it is recommended that the first dressing change does not take place in the first 24 h postoperatively, assuming the dressing is dry and sits properly (15).

Figure 2



Treatment approach in fracture-related infection. Modified from (13).

Treatment

General aspects

The aim of FRI treatment is infection-free consolidation of the fracture. This is based on a combination of surgical treatment and antibiotic therapy and should be carried out in an interdisciplinary manner. Numerous facets—from soft tissue status, local blood circulation, and underlying diseases to the advanced age of the patient, their psychological processing of the trauma, and antibiotic therapy lasting several weeks—can fall outside the expertise of orthopedic and trauma surgeons. In such cases, a number of different specialties should be involved. Joint medical rounds or boards are a proven means of closely and efficiently coordinating diagnosis and treatment in the interests of the patient and reducing revision and amputation rates, as shown in a recently published retrospective cohort study (25) *(Table 2)*.

Surgical measures

Antibiotic treatment alone is not usually sufficient for complete infection resolution due to the pathogens living in the biofilm on the implants *(Figure 2)*. A number of criteria play a role in the decision-making process regarding surgical treatment. In addition to the importance of intact soft tissue conditions, which are considered essential for infection resolution, an assessment of potential for further bone healing in terms of the reduction conditions and the stability of the osteosynthesis is important. Furthermore, there must be the possibility for sufficient debridement in order to reduce the bacterial load.

In the simplest case, fracture healing has progressed to the point where the fracture has healed. Therefore, the implant can be removed without problem and the infection can be brought under control by thorough surgical debridement of the former implant site, soft tissue, and healed bone.

In the case of acute infections involving an immature biofilm and unhealed fractures, the debridement, antibiotics, and implant retention (DAIR) approach may be considered (e15). Here, the existing osteosynthesis material is left in place, adequate surgical debridement and irrigation are performed, and antibiotic therapy, ideally local as well as systemic, is administered. The prerequisites of DAIR include sufficient soft tissue coverage, the presence of a stable implant with good reduction, and a surgically accessible implant site. For this reason, DAIR should be avoided if intramedullary interlocking nails are present, since these preclude the possibility of sufficient debridement. Moreover, leaving intramedullary nails in place is associated with a significantly higher reinfection rate (e16) *(Table 2)*. A meta-analysis based on six studies (randomized controlled trials as well as prospective and retrospective cohort studies) with a total of 276 patients showed that the DAIR procedure has the best chance of success primarily within the first 3 weeks following fracture treatment, with success rates of 86–100% (26).

In the case of implant loosening, the implant must always be removed and reosteosynthesis performed, since stability in the fracture area is a basic precondition not only for infection management but also for fracture consolidation (e17). Similarly, leaving the implant in place is no longer indicated in the case of established infection of a nonunion (e18). Therefore, in such situations, a one-stage implant replacement, a two-stage or multi-stage surgical procedure, and—in extreme cases—amputation of the affected limb represent options. If there is no bony defect and the soft tissue status is good, single-stage implant replacement with direct reosteosynthesis can be performed following adequate debridement. This is possible even if the infection has not yet been eradicated, since antibiotic therapy protects the newly inserted implant from renewed bacterial colonization. Even if plastic surgical coverage with bony defect reconstruction is required, the literature reports excellent long-term results for single-stage procedures, with 94% freedom from infection at more than 6 years (evidence level IV) (27, 28). Although it appears possible to achieve excellent results of this kind in highly specialized centers with an appropriate multidisciplinary treatment approach for limited bone defects, one must sometimes

Table 3

Targeted antibiotic treatment and oral options*1

Pathogen	Initial targeted intravenous therapy	Options for switching to the oral route
Methicillin-susceptible *Staphylococcus aureus* (MSSA) and coagulase-negative staphylococci with methicillin sensitivity	Flucloxacillin 6 × 2 g or cefazolin 3 × 2 g	If foreign material is present: rifampicin 2 × 300–450 mg plus co-trimoxazole 3 × 960 mg or levofloxacin 2 × 500 mg or doxycycline 2 × 100 mg or clindamycin 3 × 600 mg
Methicillin-resistant *Staphylococcus aureus* (MRSA) and coagulase-negative staphylococci with methicillin resistance	Vancomycin according to trough level (target: 15–20 µg/mL) or daptomycin 1× 8–10 mg/kg body weight	If foreign material is present: rifampicin 2 × 300–450 mg plus linezolid 2 × 600 mg or levofloxacin 2 × 500 mg or co-trimoxazole 3 × 960 mg or doxycycline 2 × 100 mg or clindamycin 3 × 600 mg
Streptococci	Penicillin G 6 × 4 million I.E. or penicillin G 4 × 5 million I.E. or ceftriaxone 1 × 2 g	Amoxicillin 3–4 × 1 g or clindamycin 3 × 600 mg
Enterococci	Ampicillin 6 × 2 g plus gentamicin 1 × 3 mg/kg (alternatively, ampicillin 6 × 2 g plus 2 × 2 g ceftriaxone*2) in ampicillin resistance or vancomycin according to trough level (target: 15–20 µg/mL) or daptomycin 1 × 8–10 mg/kg body weight in ampicillin resistance	Amoxicillin 3–4 × 1 g in ampicillin sensitivity (generally *Enterococcus faecalis*) Linezolid 2 × 600 mg in ampicillin resistance (generally *Enterococcus faecalis* or also vancomycin-resistant *Enterococcus* [VRE]), treatment duration limited to 4 weeks
Enterobacteriaceae (e.g., *Escherichia coli*, *Klebsiella*, *Proteus*, etc.)	Depending on antibiogram: beta-lactam antibiotics, e.g., ceftriaxone 1 × 2 g or piperacillin/tazobactam 3–4 × 4/0.5 g or meropenem 3 × 1 g	Ciprofloxacin 2 × 750 mg or levofloxacin 2 × 500 mg
Pseudomonas aeruginosa	Piperacillin/tazobactam 4 × 4/0.5 g or ceftazidime 3 × 2 g or meropenem 3 × 1 g if necessary, plus an aminoglycoside (e.g., tobramycin 1 × 5 mg/kg body weight for a short period)	Ciprofloxacin 2 × 750 mg or levofloxacin 2 × 500 mg
Gram-positive anaerobes (*Cutibacterium acnes, Finegoldia magna, Peptostreptococcus spp.*)	Ampicillin/sulbactam 3 × 2/1 g	Metronidazole 3 × 400–500 mg or clindamycin 3 × 600 mg or amoxicillin 3–4 × 1 g
Candida sp., fluconazole-sensitive	Caspofungin 1 × 50–70 mg	Fluconazole 1 × 400 mg
Culture-negative	Ampicillin/sulbactam 3 × 2/1 g	Rifampicin 2 × 300–450 mg plus levofloxacin 2 × 500 mg

I.E., international units
*1 The selection of the optimal substance and dose is made on the basis of concomitant diseases, allergies, organ dysfunction, and naturally, any resistance present. The indicated dosages apply to patients with healthy kidneys (for the purposes of conciseness, the recommendations have been modified and shortened [39]).
*2 As a possible alternative, similar to the treatment options for endocarditis (e24)

consider a two-stage or multi-stage procedure as the standard of care for soft tissue and bone defects. The aim of the treatment is to control the existing infection in the first step and to reconstruct the bony defect in the second step once the infection has eased. Programmed revisions with multiple wound irrigation procedures should be a thing of the past in view of the anesthetic burden on the patient and the risk of secondary contamination of the wound with other bacteria (e19). Depending on the clinical findings, it may be necessary in exceptional cases to repeat the debridement several times to eradicate the infection if it does not resolve, in which case this is referred to as a multi-stage procedure (evidence level III, *Table 2*).

In addition to adequate surgical debridement, the key to treatment success lies in dead space management following bone resection with antibiotic-loaded carriers as well as soft tissue management (29, 30). If there is a soft tissue defect, it is essential to achieve early soft tissue closure, as stated above. In the short term, that is, limited to a few days, vacuum assisted closure (VAC) or antibiotic bead pouches can be used (e20). Bacterial colonization of the VAC system can potentially be considered as causal in poorer treatment outcomes (30–32). If plastic surgical expertise is not available on site, prompt transfer of the patient to an appropriate center is advised.

For the reconstruction of bony defects, various reconstruction methods are available depending on the localization, size, and shape of the defect. In this context, what is referred to as the Masquelet technique has become increasingly established in recent years as a two-stage procedure. Here, the infection is initially eradicated during the first surgical procedure using spacers coated with antibiotic-containing PMMA bone cement and a well-vascularized neomembrane is formed around the bony defect zone. In a second surgery, the spacer is removed after approximately 6 weeks and the defect is filled with autologous or allogenic bone (e21). In addition, numerous bone replacement materials are available (e22). For segmental bone defects, callus distraction procedures, such as segment transport according to the Ilizarov technique, have proven their worth (e23). In 3% of cases, amputation must be taken into consideration as the best treatment option (33). This may be necessary particularly in elderly and multimorbid patients in the case of severe infection *(Table 2)*.

Antibiotic therapy

Antibiotic treatment should be initiated immediately upon completion of intraoperative specimen collection for microbiological analysis and if there is clinical suspicion of infection. An exception is made in the case of septic patients, in whom treatment should begin once blood cultures have been taken. Here, the calculated treatment should ideally cover the local spectrum of pathogens; for example, a glycopeptide antibiotic (vancomycin) can be combined with a beta-lactam (ceftriaxone or alternatively amoxicillin/clavulanic acid) to cover both the gram-positive and gram-negative spectrum. Of course, antibiotic treatment needs to be adjusted as soon as the pathogen has been detected (13).

According to current knowledge, the additive use of rifampicin is only beneficial in staphylococcal infections if foreign material is still present. In general, one waits for wound healing before adding rifampicin. Due to its high oral bioavailability, rifampicin can be administered orally from the outset. However, it is important to bear in mind that rifampicin can interact strongly with other drugs. Therefore, a review for possible interactions with the patient's concomitant medication, particularly with new oral anticoagulants (NOACs) and phenprocoumon, should be mandatory (13).

Particularly for linezolid, oral administration for longer than 4 weeks is not possible due to the bone marrow toxicity that frequently occurs. In such cases, the only remaining option is outpatient parenteral antibiotic therapy (OPAT).

No specific studies are available on the duration of treatment for FRI; the recommendations for prosthetic infections are often used as a guide, meaning that the duration of treatment is usually 12 weeks *(Table 3)*.

Conclusion

Fracture-related infection presents an increasing challenge in clinical routine. Against this backdrop, preventive measures should be optimized and treatment should always be carefully decided upon by an interdisciplinary team. The evidence for the various treatment approaches is largely based on retrospective cohort studies.

Conflict of interests statement
The authors declare that no conflict of interests exists.

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Supplementary material:
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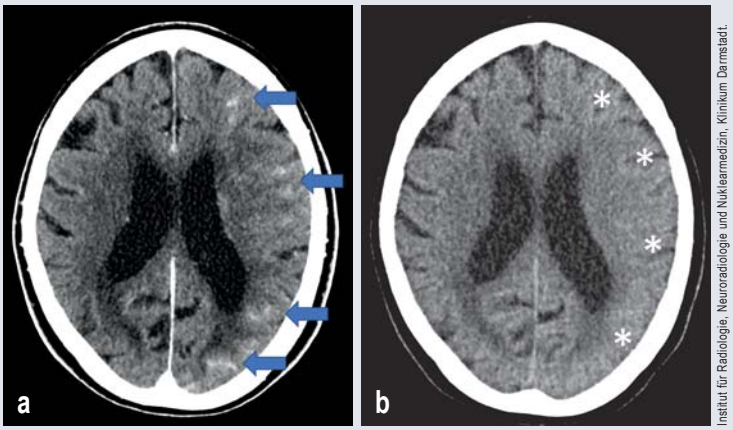
CLINICAL SNAPSHOT

Contrast-Induced Encephalopathy

An 86-year-old female patient was admitted with acute posterolateral ST elevation myocardial infarction in the setting of triple-vessel coronary artery disease. Coronary angiography was performed with 190 mL Ultravist 370. In the peri-interventional period, reduced alertness (somnolence, GCS 12), left-sided hemiparesis, and bilateral signs of pyramidal tract dysfunction (Babinski reflex) were noted. Imaging with native computed tomography on the same day enabled visualization of cortical hyperdensity and cortical edema *(Figures a and b)*. Neither could be visualized on the following day. Electroencephalography revealed severe diffuse slowing without epilepsy-typical potentials. A trial attempt at treatment with levetiracetam, an antiepileptic drug (500 mg 1–0–1), was undertaken. After 72 h, neurological symptoms were improving. We assumed the case to be one of contrast-induced encephalopathy. With an incidence of 0.3–2%, this condition is very rare. From a pathophysiological perspective, it involves direct neurotoxicity with damage to the blood–brain barrier and endothelial dysfunction. Hypertension, diabetes, kidney failure, large volumes of contrast, and male sex are considered to be risk factors. Symptoms usually resolve spontaneously and completely (after 24–72 h). Where possible, the re-use of contrast should be avoided.



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a) Prominent bilateral hyperdensity in the frontoparietal networks left > right (arrows), DDx: subarachnoid hemorrhage, DDx: contrast residue; b) signs of cortical edema, corticomedullary boundary obliteration (asterisks)

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Supplementary material to:

Fracture-Related Infection—Epidemiology, Etiology, Diagnosis, Prevention, and Treatment

by Markus Rupp*, Nike Walter*, Susanne Baertl, Robert Heyd, Florian Hitzenbichler, and Volker Alt

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Questions on the article in issue 1/2024:

CME plus+

Fracture-Related Infection—Epidemiology, Etiology, Diagnosis, Prevention, and Treatment

The submission deadline is 11 January 2025. Only one answer is possible per question. Please select the answer that is most appropriate.

Question 1

What percentage of all fractures treated on an inpatient basis in Germany in 2018 resulted in a fracture-related infection?

a) 0.23%
b) 1.23%
c) 5.32%
d) 10.13%
e) 12.30%

Question 2

Which pathogens are cited as the two most common pathogens for causing fracture-related infections?

a) *Staphylococcus aureus* and *Staphylococcus epidermidis*
b) *Streptococcus pyogenes* and *Staphylococcus haemolyticus*
c) *Escherichia coli* and *Cutibacterium* species
d) *Streptococcus pyogenes* and *Escherichia coli*
e) *Staphylococcus haemolyticus* and *Cutibacterium* species

Question 3

What is meant by sonication in the context of a fracture-related infection?

a) Reducing the bacterial count on a prosthetic implant using transcutaneous ultrasound
b) Dislodging possible bacteria from a removed implant using ultrasound for the purposes of a more precise diagnosis
c) Determining the thickness of the biofilm on a prosthetic implant by means of a transcutaneous ultrasound examination
d) The transcutaneous use of high-frequency sounds to reduce the bacterial count on an implant in the case of acute infection
e) Reducing the bacterial count on the periosteum using transcutaneous ultrasound

Question 4

Which of the following examples corresponds to the perioperative antibiotic treatment recommended in the article for closed fractures and elective procedures?

a) A single dose of doxycycline 10–20 min prior to surgery
b) A single dose of rifampicin 20–40 min prior to surgery
c) A single dose of cefazolin 15–60 min prior to surgery
d) A single dose of co-trimoxazole 60–90 min prior to surgery
e) A single dose of trimethoprim 5–15 min prior to surgery

Question 5

What percentage of fracture-related infections are caused by methicillin-resistant *Staphylococcus aureus* strains in the various countries?

a) 1% in Germany, 44% in the USA, 25% in China
b) 25% in Germany, 1% in the USA, 44 % in China
c) 44% in Germany, 25% in the USA, 1% in China
d) 14% in Germany, 24% in the USA, 31% in China
e) 31% in Germany, 1% in the USA, 25% in China

Question 6

Which of the statements on the treatment of grade I and grade II open fractures is valid?

a) Antibiotics should not be administered for longer than 24 h.
b) No antibiotics should be used.
c) Irrigation with saline solution is strongly advised against.
d) Antibiotic therapy should only be administered once an external fixator has been applied.
e) Antibiotics should be administered for at least 72 h.

Question 7

Assuming the dressing is in place and dry on the outside following fracture surgery, what is the period recommended in the text as the earliest time for the first postoperative dressing change?

a) After 6 h
b) After 8 h
c) After 12 h
d) After 24 h
e) After 48 h

Question 8

Which statement on the 'debridement, antibiotics, implant retention' (DAIR) approach is valid?

a) If carried out within the first 3 weeks following fracture treatment, it is 86–100% successful.
b) It should not be used if intramedullary interlocking nails are present.
c) It can be used even in the case of insufficient soft tissue coverage.
d) It is particularly good in the case of implant loosening.
e) DAIR is always combined with an implant replacement

Question 9

Linelozid should not be given for a prolonged period (4 weeks). What reason is given for this in the article?

a) Frequent bone marrow toxicity
b) Its reduction in efficacy due to rapid resistance development
c) Strong interactions with other drugs occur.
d) Addiction is easily developed.
e) Patients often develop an allergy to the drug.

Question 10

How does the article refer to infections originating in the skin, oral cavity, and respiratory or urinary tract that spread from the bloodstream to the fracture region via the colonization of pathogens?

a) Delocalized infections
b) Disseminated infections
c) Metastatic infections
d) Hematogenous infections
e) Dissociated infections